Uttarakhand

State E-Waste Management Report

Reporting Period: FY 2025–26 | Compiled: 20 July 2026

Semester	**T.E. Semester V – CMPN**
Subject	**E-Waste and Environmental management**
Subject Professor In-charge	**Prof. Aniket Kundu**
Assisting Teachers	**Prof. Aniket Kundu**
Laboratory	**M 315**

Name	**Pratiksha Mujmule**
Roll no	**24102A0024**
Branch	**CMPN**

1. Executive Summary

Uttarakhand's state e-waste management performance, as compiled across seven monitoring dimensions, scores 73.2 out of 100 — placed in the “Good” band (65–84). Generation inventory and formal recycling capacity utilization are the strongest areas, while informal-sector integration and collection coverage remain the most significant gaps.

State E-Waste Management Score:

73.2 / 100 **GOOD**

Key figures across the reporting sheets:

- Total e-waste generated (13 districts): 3,370,000 kg (≈3,370 tonnes), from a base population of ≈10.08 million
- Total collected: 1,463,000 kg — a state-wide collection rate of 43.4% of generation
- EPR collection target vs. achieved: 1,944,000 kg target vs. 1,291,000 kg achieved (66.4% of target)
- Installed formal recycling capacity: 27,600 tonnes/yr across 10 authorized recyclers; 16,300 tonnes processed this year (59.1% capacity utilization), at a weighted average material recovery rate of 75.4%
- EPR compliance: 6 of 10 registered producers fully Compliant (60%); 2 Partially Compliant; 1 Non-Compliant; 1 Under Review
- Informal sector: 8 known hotspots, an estimated 555 informal workers, of whom 186 (33.5%) are registered/trained under formalization programs; only 1 hotspot fully formalized
- Environmental & health monitoring: 7 hotspot sites monitored, health camps conducted at 5 (71.4%), screening 940 beneficiaries; 3 of 7 sites showed contamination exceeding safe limits

- Awareness & outreach: 8 programs conducted, reaching 8,175 participants, including 3 formal collection drives (school/RWA)

2. State Scorecard

The scorecard weights seven categories to 100 points total. Each category's achieved value is drawn directly from the corresponding data sheet totals; achievement is capped at 100% of benchmark for scoring purposes.

#	Category	Weight	Benchmark / Target	Achieved Value	Achievement %	Score Earned
1	Generation & Inventory	15	13 districts	13 districts	100.0%	15.0
2	Collection Performance	20	100%	43.4%	43.4%	8.7
3	Formal Recycling & Treatment	20	100%	1,114.1%*	100.0%	20.0
4	Regulatory Compliance	15	100%	60.0%	60.0%	9.0
5	Informal Sector Integration	10	100%	33.5%	33.5%	3.4
6	Environmental & Health Safeguards	10	100%	71.4%	71.4%	7.1
7	Awareness & Outreach	10	5,000 people	8,175 people	100.0%	10.0
	TOTAL SCORE	**100**				**73.2**

**Formal recycling achievement exceeds 100% because installed-capacity throughput (measured in tonnes, sourced from multiple regions) is not directly comparable to district-collected volume (measured in kg, local collection only); the score is capped at 100% to avoid overstating performance. This mismatch is a known limitation of the sample template, not a real operational finding.*

Score interpretation bands:

- 85–100: Excellent
- 65–84: Good ← current state score
- 45–64: Needs Improvement
- 0–44: Critical

3. Generation Inventory

District-wise e-waste generation, broken down by E-Waste (Management) Rules, 2022 Schedule I category (simplified). All figures in kilograms unless noted.

District	Population	IT & Telecom	Consumer Elec.	Large/Small Equip.	Lighting	Batteries	Other	Total Generated
Dehradun	17,00,000	4,50,000	2,25,000	1,50,000	37,500	62,500	25,000	9,50,000
Haridwar	18,90,000	3,40,000	1,70,000	1,14,000	28,000	48,000	20,000	7,20,000
Udham Singh Nagar	16,50,000	2,27,000	1,14,000	76,000	19,000	31,000	13,000	4,80,000
Nainital	9,54,000	1,80,000	90,000	60,000	15,000	25,000	10,000	3,80,000
Almora	6,22,000	71,000	35,500	23,500	6,000	9,500	4,500	1,50,000

District	Population	IT & Telecom	Consumer Elec.	Large/Small Equip.	Lighting	Batteries	Other	Total Generated
Pauri Garhwal	6,87,000	66,000	33,000	22,000	5,500	9,000	4,500	1,40,000
Tehri Garhwal	6,18,000	61,500	30,500	20,500	5,000	8,500	4,000	1,30,000
Uttarkashi	3,30,000	40,000	20,000	13,500	3,500	5,500	2,500	85,000
Pithoragarh	4,83,000	45,000	22,500	15,000	3,800	6,200	2,500	95,000
Chamoli	3,91,000	42,500	21,500	14,000	3,500	6,000	2,500	90,000
Champawat	2,59,000	26,000	13,000	8,500	2,200	3,800	1,500	55,000
Bageshwar	2,49,000	23,500	12,000	7,800	2,000	3,400	1,300	50,000
Rudraprayag	2,42,000	21,000	10,800	7,000	1,800	3,100	1,300	45,000
STATE TOTAL	**1,00,75,000**	**15,93,500**	**7,97,800**	**5,31,800**	**1,32,800**	**2,21,500**	**92,600**	**33,70,000**

IT & Telecom equipment (computers, mobiles) is the largest category statewide at 1,593,500 kg (47.3% of total generation), consistent with high device turnover in this category nationally. Dehradun and Haridwar together account for just under half of total state generation, reflecting their larger, more urbanized populations.

4. Collection Performance

'Generated' is pulled from the Generation Inventory. 'EPR Collection Target/Achieved' reflect producer-responsibility obligations tracked separately from general collection. All figures in kilograms.

District	Generated	Collected	Reg. Centers	EPR Target	EPR Achieved
Dehradun	9,50,000	4,70,000	22	5,70,000	4,10,000
Haridwar	7,20,000	3,30,000	18	4,32,000	3,00,000
Udham Singh Nagar	4,80,000	2,10,000	15	2,88,000	1,90,000
Nainital	3,80,000	1,65,000	12	1,50,000	1,38,000
Almora	1,50,000	58,000	6	90,000	50,000
Pauri Garhwal	1,40,000	52,000	5	84,000	46,000
Tehri Garhwal	1,30,000	47,000	5	78,000	41,000
Uttarkashi	85,000	27,000	3	51,000	24,000
Pithoragarh	95,000	30,000	4	57,000	27,000
Chamoli	90,000	29,000	4	54,000	26,000
Champawat	55,000	17,000	3	33,000	15,000
Bageshwar	50,000	15,000	3	30,000	13,000
Rudraprayag	45,000	13,000	2	27,000	11,000
STATE TOTAL	**33,70,000**	**14,63,000**	**102**	**19,44,000**	**12,91,000**

Collection coverage varies widely by geography: plains districts (Dehradun, Haridwar, Udham Singh Nagar, Nainital) report stronger formal collection infrastructure, while hill districts (Uttarkashi, Bageshwar, Rudraprayag) show the lowest registered-center density, consistent with the logistical challenges of collection in mountainous terrain.

5. Recycling Infrastructure

Authorized recyclers/dismantlers currently operating in the state. Capacity and quantity processed are in tonnes/year.

Recycler / Dismantler	District	Installed Capacity	Processed This Year	Recovery Rate
GreenTech E-Waste Recyclers Pvt. Ltd.	Dehradun	6,000	4,100	82.0%
Doon Valley Dismantlers Pvt. Ltd.	Dehradun	3,000	1,850	76.0%
Ganga Recyclers & Refiners Pvt. Ltd.	Haridwar	5,500	3,200	79.0%
Haridwar Metal Recovery Co.	Haridwar	2,800	1,650	71.0%
Terai E-Waste Solutions Pvt. Ltd.	Udham Singh Nagar	3,200	1,900	74.0%
Kumaon Green Recyclers Pvt. Ltd.	Nainital	1,500	820	68.0%
Himalaya Circular Systems Pvt. Ltd.	Udham Singh Nagar	2,200	1,100	70.0%
Rishikesh E-Scrap Processors	Dehradun	1,800	950	65.0%
Bhagirathi Recyclers Pvt. Ltd.	Uttarkashi	400	120	55.0%
Sarovar Metals & Recovery Pvt. Ltd.	Nainital	1,200	610	66.0%
STATE TOTAL	**—**	**27,600**	**16,300**	**75.4%**

Formal recycling capacity is concentrated in Dehradun and Haridwar (5 of 10 authorized facilities), which also host the largest single-site capacities. The state-wide weighted average material recovery rate of 75.4% is led by GreenTech E-Waste Recyclers (Dehradun, 82.0%) and lagged by Bhagirathi Recyclers (Uttarkashi, 55.0%), the only authorized facility in a hill district in this sample.

6. Regulatory Compliance (EPR)

Compliance status of registered producers/EPR entities against their Extended Producer Responsibility obligations.

Producer / EPR Entity	Compliance Status	Inspection Planned	Inspection Conducted	Penalty Issued	Penalty Resolved
Sunrise Electronics Ltd.	Compliant	Yes	Yes	No	N/A
Himshakti Appliances Pvt. Ltd.	Compliant	Yes	Yes	No	N/A
Doon Devices Pvt. Ltd.	Partially Compliant	Yes	Yes	Yes	No
Kumaon Gadgets Ltd.	Compliant	Yes	No	No	N/A
Terai Home Systems Pvt. Ltd.	Non-Compliant	Yes	Yes	Yes	No
Ganga Consumer Tech Ltd.	Compliant	Yes	Yes	No	N/A
Rishikesh Electricals Pvt. Ltd.	Under Review	No	No	No	N/A
Bhagirathi Electronics Ltd.	Compliant	Yes	Yes	No	N/A
Nanda Devi Appliances Pvt. Ltd.	Partially Compliant	Yes	Yes	Yes	Yes
Vasundhara Tech Industries Ltd.	Compliant	Yes	Yes	No	N/A

Metric	Count
Compliant	6
Partially Compliant	2
Non-Compliant	1
Under Review	1
Total Registered	10

Metric	Count
Inspections Planned	9
Inspections Conducted	8
Penalties Issued	3
Penalties Resolved	1

60% full compliance leaves room for improvement; the one Non-Compliant entity (Terai Home Systems, below 40% of its EPR target) and two Partially Compliant entities account for the bulk of the gap. Of the 3 penalties issued state-wide, only 1 has been resolved to date.

7. Informal Sector Hotspots

Known informal e-waste recycling clusters and their formalization status.

Hotspot / Cluster	District	Status	Est. Informal Workers	Workers Registered/Trained
Patel Nagar Scrap Yard	Dehradun	Partially Formalized	90	35
Roorkee Scrap Market	Haridwar	Informal	120	20
Kashipur Scrap Cluster	Udham Singh Nagar	Partially Formalized	75	28
Haldwani Old Bazaar Yard	Nainital	Informal	60	10
Rudrapur Industrial Scrap Belt	Udham Singh Nagar	Formalized	50	50
Vikasnagar Scrap Point	Dehradun	Informal	40	5
Jwalapur Recycling Lane	Haridwar	Partially Formalized	85	30
Ramnagar Scrap Cluster	Nainital	Informal	35	8

Of an estimated 555 informal workers across 8 hotspots, 186 (33.5%) have been registered or trained under formalization programs; only Rudrapur Industrial Scrap Belt is fully formalized. This is the lowest-scoring category in the scorecard and represents the largest opportunity for targeted intervention.

8. Environmental & Health Monitoring

Environmental testing and health camps conducted at informal-sector hotspots.

Site Monitored	District	Monitoring Type	Result	Health Camp	Beneficiaries
Patel Nagar Scrap Yard	Dehradun	Soil + Water	Exceeds limit (Pb, Cd)	Yes	210
Roorkee Scrap Market	Haridwar	Air Quality	Within limit	Yes	180
Kashipur Scrap Cluster	Udham Singh Nagar	Soil	Exceeds limit (Pb)	Yes	150
Haldwani Old Bazaar Yard	Nainital	Water	Within limit	No	0
Rudrapur Industrial Scrap Belt	Udham Singh Nagar	Soil + Water	Within limit	Yes	260
Jwalapur Recycling Lane	Haridwar	Air Quality	Exceeds limit (particulate)	Yes	140
Vikasnagar Scrap Point	Dehradun	Soil	Within limit	No	0

3 of 7 monitored sites (Patel Nagar, Kashipur, Jwalapur) showed contamination exceeding safe limits — lead/cadmium in soil or water at two sites and particulate air pollution at one. Health camps have been conducted at 5 of 7 sites, screening 940 beneficiaries in total; camps remain pending at the two lowest-activity hotspots.

9. Awareness & Outreach Programs

Public awareness campaigns, school/RWA collection drives, and retailer sensitization programs conducted.

Program Name	District	Type	Participants Reached
E-Waste Awareness Drive – Schools	Dehradun	School Collection Drive	1,200
Haridwar Kumbh Cleanup & Awareness	Haridwar	Public Campaign	3,500
RWA Take-Back Scheme Promotion	Udham Singh Nagar	RWA Collection Drive	850
College E-Waste Pledge Campaign	Nainital	Institutional Drive	600
Rural Outreach – Hill Districts	Almora	Community Workshop	320
Retailer Take-Back Training	Dehradun	Retailer Sensitization	95
Uttarkashi School Collection Drive	Uttarkashi	School Collection Drive	210
Digital India Week E-Waste Booth	Haridwar	Public Campaign	1,400

8 programs reached 8,175 participants against a 5,000-person benchmark (163% of target), the strongest over-performance in the scorecard. Public campaigns (Haridwar Kumbh Cleanup, Digital India Week) drove the largest reach; only 3 of the 8 programs were structured as formal collection drives.

10. Key Observations & Recommendations

Based on the patterns in this sample dataset:

- Close the collection gap: at 43.4% of generation, over half of e-waste generated is not entering the formal collection stream. Expanding registered collection centers in under-served hill districts (Uttarkashi, Bageshwar, Rudraprayag) would likely have the highest marginal impact.
- Prioritize informal-sector formalization: with only 33.5% of estimated informal workers registered or trained, this is the lowest-scoring category. Scaling the Rudrapur ‘model formalization’ approach to other hotspots is a concrete next step.
- Follow up on non-compliance and unresolved penalties: 1 producer is Non-Compliant and only 1 of 3 issued penalties has been resolved; timely resolution strengthens EPR enforcement credibility.
- Extend environmental health monitoring: 2 of 7 hotspots have no health camp scheduled despite being known informal clusters; closing this gap protects worker and community health.
- Sustain awareness momentum: outreach exceeded its participant benchmark; converting more general-awareness events into structured collection drives (currently only 3 of 8 programs) could translate awareness into measurable collection volume.

Data Sources & Methodology Note

> *All figures in this report originate from a single illustrative Excel workbook (8 linked sheets: Scorecard, Generation Inventory, Collection Log, Recycling Infrastructure, Compliance Tracker, Informal Sector Hotspots, Env. Health Monitoring, Awareness Programs) built for template and demonstration purposes. No data in this report has been sourced from CPCB, Uttarakhand SPCB, or any government publication. For an official report, replace every figure with verified data from the Uttarakhand State Pollution Control Board and CPCB's E-Waste portal.*